

Uthara A S

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PROFILE

Data Scientist and researcher with expertise in scientific machine learning, time-series analysis, and signal processing. Skilled in developing deep learning models. Experience working with scientific and spatiotemporal datasets, with applications in space data science and Earth observation. Strong foundation in computational methods and research-oriented workflows. Seeking PhD opportunities in machine learning for space weather, heliophysics.

EXPERIENCE

Research Collaborator, ISRO

March 2026 – Present

- Detection of Solar Wind Switchbacks using 1D-CNN, BiLSTM and Transformer-based Models on In-situ Magnetic Field and Plasma Time-Series Data of NASA Parker Solar Probe.
- Building end-to-end signal processing pipelines: noise filtering, feature extraction, and temporal segmentation for classification of solar wind events.
- Evaluated model predictions against expert-labelled switchback intervals, identifying performance gaps and improving detection accuracy through iterative model refinement.

Probationary Officer, Federal Bank

July 2025 – March 2026

- Contributed to real-world operational workflows by analyzing processes and supporting data-driven decision-making initiatives.
- Collaborated with cross-functional teams to identify issues, propose solutions.

Research Intern, IITK

May 2024-July 2024

- Designed a hybrid Transfer Learning + Decision Tree model for financial fraud detection on imbalanced data; improved classification accuracy from 93.8% (baseline) to 94% published in IEEE.
- Engineered deep learning feature extraction pipelines combined with interpretable decision tree classifiers, outperforming traditional methods on precision, recall, and generalization across large-scale transaction datasets.

EDUCATION

• MSc Data Science with Logistics & Supply Chain Management

CGPA 8.8/10 | 2023-25

Amrita Vishwa Vidyapeetham

Thesis: Developed a multi-objective company performance scoring system (CTG-S) using Adaptive Discriminant Analysis (ADA), PCA and KMeans clustering — validated via Six Sigma quality metrics

• Bachelor of Computer Application in Data Science

CGPA 8.5/10 | 2020-23

Amrita Vishwa Vidyapeetham

SKILLS

- **Programming:** Python, R, SQL, MATLAB, Linux, Git/GitHub
- **ML / Deep Learning:** PyTorch, TensorFlow, Scikit-learn, 1D-CNN, Transformer Models, BiLSTM, Transfer Learning
- **Scientific Computing:** NumPy, SciPy, Pandas, Time-Series Analysis, Signal Processing, Statistical Modeling
- **Soft Skills:** Research Orientation, Analytical Thinking, Scientific Communication, Problem Solving

PUBLICATIONS & CERTIFICATIONS

- Financial Fraud Detection Using an Advanced Hybrid Model, **IEEE** -[Publication Link](#)
- Customer Segmentation Using RFM Model and a New Approach to Encourage Loyal Customers, **Springer Nature** Singapore, 2024-[Publication Link](#)
- IBM (Data Engineering, Data Privacy), Google (Data Analytics, Analytics IQ, IITK Research Internship